

**Mid Term 2**  
**Digital System Design (CSIC-100)**

**Time:** 50 minutes

**Max Marks:-15**

All questions are compulsory

**Ques 1.** Design a 40:1 Mux using an 8:1 Mux only.

(4 marks)

**Ques 2.** Implement the function  $A \oplus B \oplus C$  using a 4:1 Mux

(2 marks)

**Ques 3.** Use the tabular method to obtain the minimal expression for

$$F = \sum m(6,7,8,9) \cdot d(10,11,12,13,14,15)$$

(7 marks)

**Ques4** Design a 2: 4 priority encoder

(2 marks)